$P$ : Probability $\equiv$ volume of clay

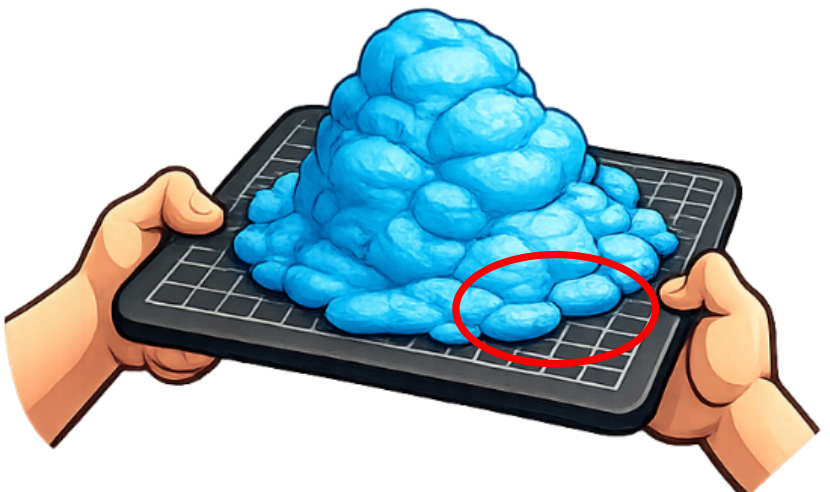
$\left|\det \boldsymbol{J}_f\right|$ : Jacobian $\equiv$ strech factor

$d\boldsymbol{x}$ or $d\boldsymbol{z}$ : measure $\equiv$ grid on the sheet

$f$ : transformation $\equiv$ strech

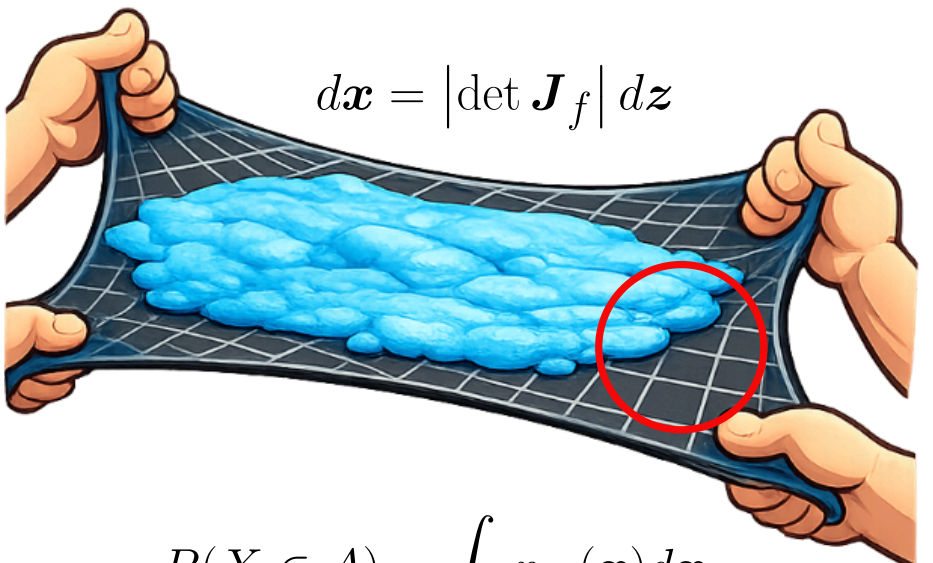
$A$ : measurement set $\equiv$ red circle

$p(\cdot)$ : $PDF \equiv$ thickness of clay



$$P(Z \in f^{-1}(A)) = \int_{f^{-1}(A)} p_Z(\boldsymbol{z}) d\boldsymbol{z}$$

$$d\boldsymbol{x} = \left|\det \boldsymbol{J}_f\right| d\boldsymbol{z}$$



$$P(X \in A) = \int_{A} p_X(\boldsymbol{x}) d\boldsymbol{x}$$